

2026-06-08

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1

- **A** — : “ ” (Bellman–Ford, Dijkstra).
- **B** — : “ ”.

B . A , B first-principles .

i

$$D_{ij}(F_{ij}) = \frac{F_{ij}}{C_{ij} - F_{ij}} \quad \text{M/M/1} \quad \rho/(1-\rho) \quad \cdot \quad \sum_{(i,j)} D_{ij} \quad \text{Little}$$

2 A.

2.1 Optimality Principle Sink Tree

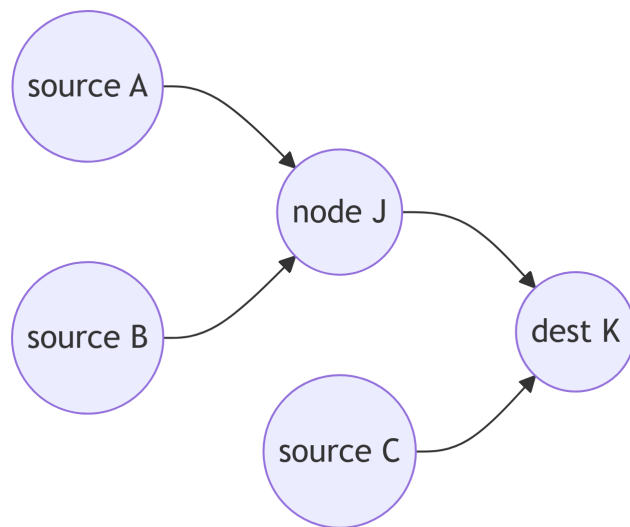
$J \rightarrow I \rightarrow K$, $J \rightarrow K$.

: K K (sink tree) .
(). sink tree .

(Bellman) , Bellman-Ford .

graph LR

```
S1((source A)) --> J((node J))
S2((source B)) --> J
J --> K((dest K))
S3((source C)) --> K
```



2.2 Distance Vector = Bellman-Ford

(,) . 1 :

- $d_{ij} = \infty$: (i, j) arc
- D_i^h : i 1 h arc
- $D_1^h = 0$ ($\forall h$), $D_i^0 = \infty$ ($\forall i \neq 1$)

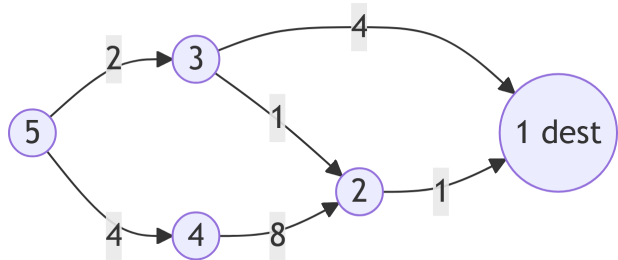
$$D_i^{h+1} = \min_j [d_{ij} + D_j^h], \quad \forall i \neq 1$$

: “ hop j $(d_{ij}) +$ h hop (D_j^h) ” . hop DP , $h = N - 1$.

2.2.1

1. 1 .

```
graph LR
  N2((2)) -->|1| N1((1 dest))
  N3((3)) -->|4| N1
  N3 -->|1| N2
  N4((4)) -->|8| N2
  N5((5)) -->|2| N3
  N5 -->|4| N4
```



```
import math
import pandas as pd

nodes = [1, 2, 3, 4, 5]
# (1) out-arc: i -> (j, d_ij)
arcs = {1: [], 2: [(1, 1)], 3: [(1, 4), (2, 1)], 4: [(2, 8)], 5: [(3, 2), (4, 4)]}

INF = math.inf
D = {i: (0 if i == 1 else INF) for i in nodes}
history = [dict(h=0, **{f"D{i}": D[i] for i in nodes})]

for h in range(1, 5):
    newD = {i: (0 if i == 1 else INF) for i in nodes}
    for i in nodes:
        if i == 1:
            continue
        cand = [d_ij + D[j] for (j, d_ij) in arcs[i]]
        newD[i] = min(cand) if cand else INF
    D = newD
    history.append(dict(h=h, **{f"D{i}": D[i] for i in nodes}))

pd.DataFrame(history).set_index("h")
```

$h = 1$, $h = 2$ 3 2 4 $\rightarrow$ 2 , $h = 3$ 5 3 6 $\rightarrow$ 4 $h = 3$. “ ” hop .

2.2.2 Count-to-Infinity

() () . $A - B$ C (A stale) “ B 2 ” , A “ C 3 ” .
 C “ 4 ” ... 1 ∞ .

	D1	D2	D3	D4	D5
h					
0	0	inf	inf	inf	inf
1	0	1.0	4.0	inf	inf
2	0	1.0	2.0	9.0	6.0
3	0	1.0	2.0	9.0	4.0
4	0	1.0	2.0	9.0	4.0

Figure 1

! DV

1. (count-to-infinity): hop
2. : $\rightarrow$ B

2.3 Link State = Dijkstra

(1), (2), (3), (4) **broadcast**, (5) **Dijkstra**.

- : $P = \{1\}$, $D_1 = 0$, $D_j = d_{j1}$ ($j \neq 1$)
- Step 1: P $D_i = \min_{j \notin P} D_j + d_{ji}$ P .
- Step 2: P j $D_j := \min [D_j, d_{ji} + D_i]$. Step 1.

greedy : arc, P i i ($\rightarrow$). **DV**
hop DP.

Distance Vector (BF)	Link State (Dijkstra)
,	,
, count-to-infinity	,
hop	

2.4

- **Hierarchical routing**: CPU $\rightarrow$.
- **Flooding**: (a) hop counter 0, (b) DB.
- **Multicast**: (join/leave), spanning tree : DVMRP, Core Based Tree.

3 B.

3.1

: $F_{ij}(\text{flow})$, $C_{ij}(\text{cost})$, $d_{ij}(\text{delay})$, $D_{ij}(\text{distance})$.

$$D_{ij}(F_{ij}) = \frac{F_{ij}}{C_{ij} - F_{ij}} + d_{ij}F_{ij}$$

M/M/1 ($\rho/(1-\rho)$, $\rho = F_{ij}/C_{ij}$), Little “”($d_{ij} \times F_{ij}$), $\gamma = \sum_w r_w$

: OD pair W, w P_w , flow x_p , flow r_w .

$$\sum_{p \in P_w} x_p = r_w \quad (\forall w), \quad x_p \geq 0, \quad F_{ij} = \sum_{p \ni (i,j)} x_p$$

:

$$\min_{\{x_p\}} D(x) = \sum_{(i,j)} D_{ij} \left(\sum_{p \ni (i,j)} x_p \right) \quad \text{s.t.} \quad \sum_{p \in P_w} x_p = r_w, \quad x_p \geq 0$$

: (1) flow, (2), (3) (splitting). A.

3.1.1

```
import numpy as np
import matplotlib.pyplot as plt

C = 1.0
F = np.linspace(0, 0.98 * C, 400)
D = F / (C - F)
Dp = C / (C - F) ** 2

fig, ax = plt.subplots(1, 2, figsize=(10, 3.8))
ax[0].plot(F / C, D, lw=2)
ax[0].set(xlabel=r"utilization $\rho=F/C$", ylabel=r"$D(F)=F/(C-F)$",
          title="Link cost (mean # in system)")
ax[0].grid(alpha=0.3)
ax[1].plot(F / C, Dp, lw=2, color="C3")
ax[1].set(xlabel=r"utilization $\rho=F/C$", ylabel=r"$D'(F)=C/(C-F)^2$",
          title="Marginal delay (per-link DFDL term)")
ax[1].grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

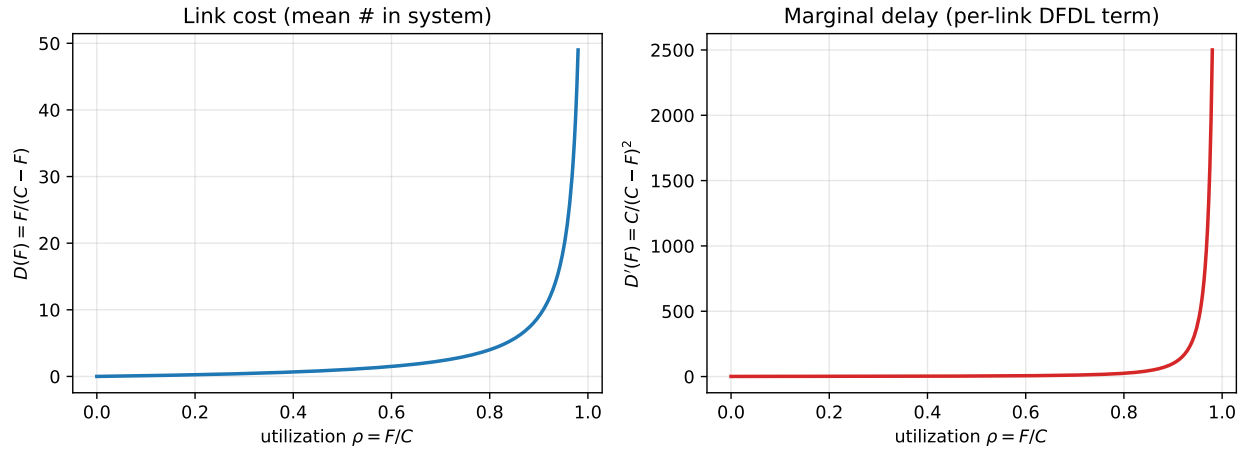


Figure 2: M/M/1 $D(F)$ $D'(F)$. 1.

3.2

p First Derivative Length (DFDL, 1):

$$\frac{\partial D(x)}{\partial x_p} = \sum_{(i,j) \in p} D'_{ij}(F_{ij})$$

“ p flow”, (marginal delay) .

: x^* OD pair p flow δ p' . 1

$$\delta \frac{\partial D(x^*)}{\partial x_{p'}} - \delta \frac{\partial D(x^*)}{\partial x_p} \geq 0.$$

!

OD pair w flow $p \in P_w$

$$x_p^* > 0 \implies \frac{\partial D(x^*)}{\partial x_p} \leq \frac{\partial D(x^*)}{\partial x_{p'}}, \quad \forall p' \in P_w.$$

flow (DFDL), flow 0 DFDL .

: , (system optimal) . () — , Wardrop , KKT .

3.3 Feasible / Descent Direction

x Δx :

- **Feasible:** $x + \alpha \Delta x$ $\alpha > 0 \rightarrow$ OD pair $\sum_{p \in P_w} \Delta x_p = 0$, $x_p = 0 \rightarrow \Delta x_p \geq 0$.
- **Descent:**

$$\sum_{w \in W} \sum_{p \in P_w} \frac{\partial D(x)}{\partial x_p} \Delta x_p < 0, \quad \frac{\partial D(x)}{\partial x_p} = \sum_{(i,j) \text{ on } p} D'_{ij}(F_{ij}).$$

step size $\alpha > 0$ $D(x + \alpha \Delta x) < D(x)$. 1 “feasible + descent + step size” .

3.4 Frank–Wolfe (Flow Deviation)

$x = \{x_p\}$:

1. OD pair MFDL(Min First Derivative Length) (D'_{ij} Dijkstra).
2. MFDL $\bar{x} = \{\bar{x}_p\}$.
3. : $\alpha^* \in [0, 1] : D[x + \alpha^*(\bar{x} - x)] = \min_{\alpha \in [0, 1]} D[x + \alpha(\bar{x} - x)]$.
4. : $x_p := x_p + \alpha^*(\bar{x}_p - x_p)$ ($\forall p$).

: D 1 , (simplex) (“ MFDL ”) . α^* .

i

MFDL α^* (equal-ratio shift) .

3.4.1

$$D(x) = \frac{1}{2}(x_1^2 + x_2^2 + 0.1x_3^2) + 0.55x_3, \quad x_1 + x_2 + x_3 = 1, \quad x_i \geq 0$$

$$\frac{\partial D}{\partial x_1} = x_1, \quad \frac{\partial D}{\partial x_2} = x_2, \quad \frac{\partial D}{\partial x_3} = 0.1x_3 + 0.55$$

$$x_3^* = 0, \quad x_1^* = x_2^* = \frac{1}{2}, \quad .$$

3.5

— **steepest descent** ($\nabla^2 f$):

$$x^{k+1} = x^k - \alpha^k \nabla f(x^k), \quad \frac{f(x^{k+1}) - f^*}{f(x^k) - f^*} \leq \left(\frac{M - m}{M + m} \right)^2$$

M, m Hessian $\cdot$ M/m ($\cdot$) B^k :

$$x^{k+1} = x^k - \alpha^k B^k \nabla f(x^k)$$

$B^k = [\nabla^2 f(x^k)]^{-1}$ **Newton's method**($\cdot$). Hessian:

$$x_i^{k+1} = x_i^k - \alpha^k \left[\frac{\partial^2 f(x^k)}{\partial x_i^2} \right]^{-1} \frac{\partial f(x^k)}{\partial x_i}$$

Positive orthant ($x \geq 0$): 0 projection $[\cdot]^+ = \max\{0, \cdot\}$:

$$x_i^{k+1} = \max \left\{ 0, x_i^k - \alpha^k \left[\frac{\partial^2 f(x^k)}{\partial x_i^2} \right]^{-1} \frac{\partial f(x^k)}{\partial x_i} \right\}$$

3.6 Gradient Projection

: $\sum_p x_p = r_w$ **MF DL** p_w **flow**.

$$x_{p_w} = r_w - \sum_{\substack{p \in P_w \\ p \neq p_w}} x_p$$

flow + positive orthant.

! flow

$$x_p^{k+1} = \max \left\{ 0, x_p^k - \alpha^k H_p^{-1} (d_p - d_{p_w}) \right\}$$

$$d_p = \sum_{(i,j) \in p} D'_{ij}(F_{ij}), \quad d_{p_w} = \sum_{(i,j) \in p_w} D'_{ij}(F_{ij}), \quad H_p = \sum_{(i,j) \in L_p} D''_{ij}(F_{ij})$$

$$L_p = p \setminus p_w \quad (\cdot). \quad \text{flow}.$$

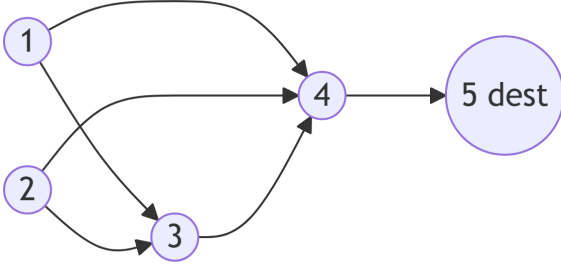
: $(d_p - d_{p_w})$ “ ”, flow H_p^{-1} (2) step. FW (unequal shift) $\rightarrow$.

3.6.1

```

graph LR
  N1((1)) --> N3((3))
  N1 --> N4((4))
  N2((2)) --> N3
  N2 --> N4
  N3 --> N4
  N4 --> N5((5 dest))

```



OD (1,5): $r_1 = 4$, OD (2,5): $r_2 = 8$. $D_{ij}(F_{ij}) = \frac{1}{2}F_{ij}^2 \rightarrow D'_{ij} = F_{ij}$, $D''_{ij} = 1$.

$p_1(1) = \{1, 4, 5\}$, $p_2(1) = \{1, 3, 4, 5\}$ / $p_1(2) = \{2, 4, 5\}$, $p_2(2) = \{2, 3, 4, 5\}$. flow $F_{13} = 4$, $F_{14} = 0$, $F_{23} = 8$, $F_{24} = 0$, $F_{34} = 12$, $F_{45} = 12$.

$$d_{p_1(1)} = F_{14} + F_{45} = 0 + 12 = 12, \quad d_{p_2(1)} = F_{13} + F_{34} + F_{45} = 4 + 12 + 12 = 28$$

MFDL $p_1(1)(=12)$. flow $p_2(1) \quad 4$. $L_p = \{(1, 3), (3, 4)\} \cup \{(1, 4)\} = \{(1, 3), (3, 4), (1, 4)\} \rightarrow H_p = 1 + 1 + 1 = 3$.

$$x_{p_2(1)} := \max \left\{ 0, 4 - \frac{\alpha^k}{3}(28 - 12) \right\} = \max \left\{ 0, 4 - \frac{16\alpha^k}{3} \right\}, \quad x_{p_1(1)} := r_1 - x_{p_2(1)}$$

3.6.2 FW vs Gradient Projection

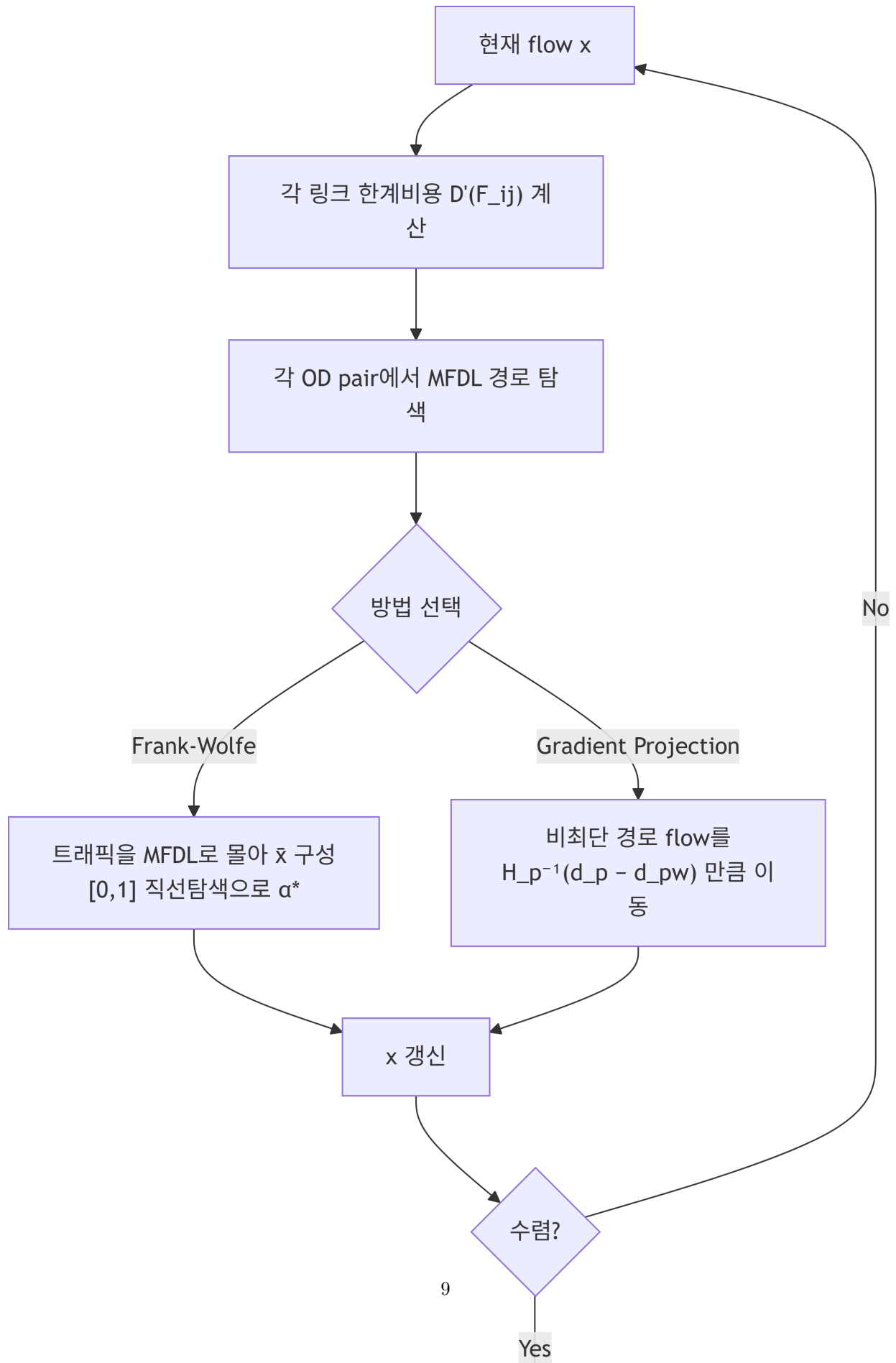
	Frank-Wolfe	Gradient Projection
flow	1 (DFDL) MFDL () (LP+)	1 · 2 (H_p) (Hessian)

3.6.3

```

flowchart TD
  A[" flow x"] --> B[" D'(F_ij) "]
  B --> C[" OD pair MFDL "]
  C --> Dn[" "]
  Dn --> E[" MFDL x̄ <br/>[0,1] *"]
  Dn --> F[" Gradient Projection" | F[" flow <br/>H_p^{-1}(d_p - d_pw) "]
  E --> G["x "]
  F --> G
  G --> H[" ?"]
  H --> I["No" | A
  H --> J["Yes" | I[" "]

```

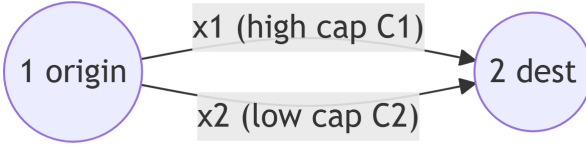



3.7 Two-Link : closed form

OD , (C_1 , C_2), r .

graph LR

```
0((1 origin)) -->|"x1 (high cap C1)"| Dst((2 dest))
0 -->|"x2 (low cap C2)"| Dst
```



$$D(x) = D_1(x_1) + D_2(x_2), \quad D_i(x_i) = \frac{x_i}{C_i - x_i}, \quad x_1^* + x_2^* = r, \quad x_i^* \geq 0$$

():

$$D'_i(x_i) = \frac{d}{dx_i} \frac{x_i}{C_i - x_i} = \frac{(C_i - x_i) + x_i}{(C_i - x_i)^2} = \frac{C_i}{(C_i - x_i)^2}$$

Case 1 — $(x_1^* = r, x_2^* = 0)$: (1) DFDL (2) DFDL:

$$\frac{C_1}{(C_1 - r)^2} \leq \frac{C_2}{C_2^2} = \frac{1}{C_2} \Rightarrow C_1 C_2 \leq (C_1 - r)^2 \Rightarrow \boxed{r \leq C_1 - \sqrt{C_1 C_2}}$$



$$r \leq C_1 - \sqrt{C_1 C_2} \quad . \quad 1/C_2, \quad r \leq r^* = C_1 - \sqrt{C_1 C_2}$$

Case 2 — $(x_1^*, x_2^* > 0)$: DFDL .

$$\frac{C_1}{(C_1 - x_1^*)^2} = \frac{C_2}{(C_2 - x_2^*)^2} \Rightarrow \frac{\sqrt{C_1}}{C_1 - x_1^*} = \frac{\sqrt{C_2}}{C_2 - x_2^*}$$

$$x_2^* = r - x_1^* \quad , \quad x_1^* :$$

$$x_1^* (\sqrt{C_1} + \sqrt{C_2}) = \sqrt{C_1} [r - (C_2 - \sqrt{C_1 C_2})]$$

$$\boxed{x_1^* = \frac{\sqrt{C_1} [r - (C_2 - \sqrt{C_1 C_2})]}{\sqrt{C_1} + \sqrt{C_2}}, \quad x_2^* = \frac{\sqrt{C_2} [r - (C_1 - \sqrt{C_1 C_2})]}{\sqrt{C_1} + \sqrt{C_2}}} \quad (r > C_1 - \sqrt{C_1 C_2})$$

($r = r^* \quad x_2^* = 0, \quad x_1^* = r^* \quad \text{Case 1} \quad .$)

3.7.1

```
import numpy as np
import matplotlib.pyplot as plt

C1, C2 = 10.0, 4.0
r_star = C1 - np.sqrt(C1 * C2)

def split(r):
```

```

if r <= r_star:
    return r, 0.0
s1, s2 = np.sqrt(C1), np.sqrt(C2)
x1 = s1 * (r - (C2 - np.sqrt(C1 * C2))) / (s1 + s2)
x2 = s2 * (r - (C1 - np.sqrt(C1 * C2))) / (s1 + s2)
return x1, x2

rs = np.linspace(0, 0.95 * (C1 + C2), 400)
X1 = np.array([split(r)[0] for r in rs])
X2 = np.array([split(r)[1] for r in rs])

plt.figure(figsize=(7, 4.2))
plt.plot(rs, X1, lw=2, label=r"$x_1^*$ (high cap $C_1=10$)")
plt.plot(rs, X2, lw=2, label=r"$x_2^*$ (low cap $C_2=4$)")
plt.axvline(r_star, ls="--", color="gray")
plt.text(r_star + 0.15, 8.4, rf"$r^*={r_star:.2f}$", color="gray")
plt.xlabel("total demand r")
plt.ylabel("optimal path flow")
plt.title("Two-link optimal routing")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```

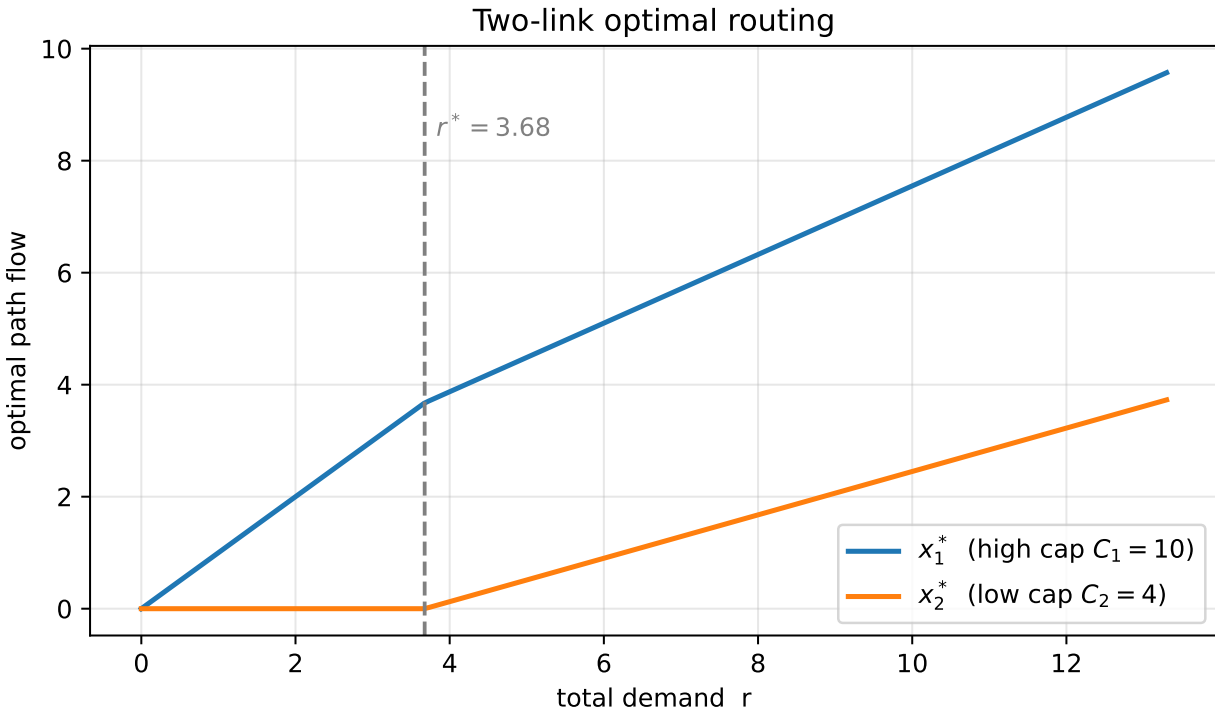


Figure 3: . r r^* .

3.7.2

```
import numpy as np
import matplotlib.pyplot as plt

C1, C2, r = 10.0, 4.0, 6.0
x1 = np.linspace(r - C2 + 1e-3, min(r, C1) - 1e-3, 400)
x2 = r - x1
Dtot = x1 / (C1 - x1) + x2 / (C2 - x2)

s1, s2 = np.sqrt(C1), np.sqrt(C2)
x1_opt = s1 * (r - (C2 - np.sqrt(C1 * C2))) / (s1 + s2)
#
g1 = C1 / (C1 - x1_opt) ** 2
g2 = C2 / (C2 - (r - x1_opt)) ** 2
print(f"x1* = {x1_opt:.4f}, x2* = {r - x1_opt:.4f}")
print(f"DFDL link1 = {g1:.5f}, DFDL link2 = {g2:.5f} (    )")

plt.figure(figsize=(7, 4.2))
plt.plot(x1, Dtot, lw=2)
plt.axvline(x1_opt, ls="--", color="C3", label=rf"$x_1^*={x1_opt:.3f}$")
plt.xlabel(r"$x_1$ (with $x_2 = r - x_1$)")
plt.ylabel(r"total cost $D(x_1)+D(x_2)$")
plt.title(f"Total delay vs split at r={r}")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

x1* = 5.0994, x2* = 0.9006
DFDL link1 = 0.41639, DFDL link2 = 0.41639 (    )
```

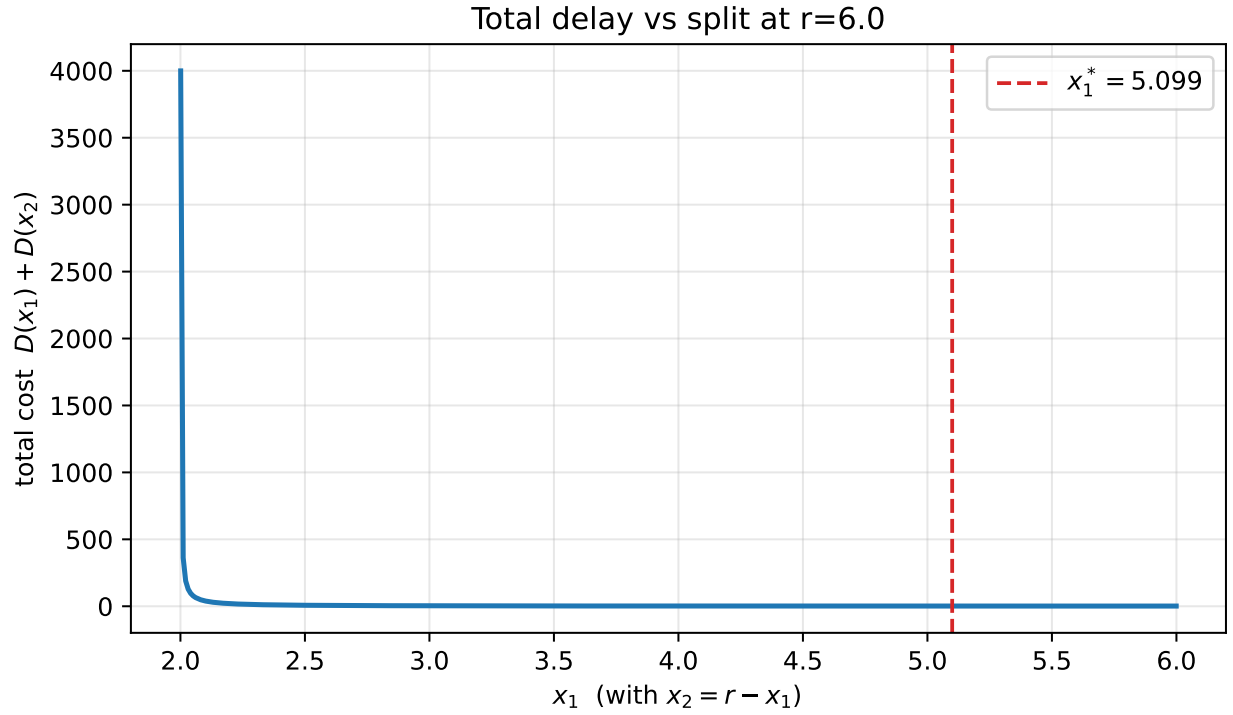


Figure 4: $r = 6.0$. The optimal split is $x_1^* = 5.099$.



(3) DFDL $\begin{cases} D_p(x_p) = \frac{x_p}{C_p - x_p} \\ \sum_p x_p = r \end{cases}$. (1) $D'_p(x_p) = \frac{C_p}{(C_p - x_p)^2}$ $\rightarrow$ (2) “ ” case $\rightarrow$.
 $C_1 > C_2 > C_3$ r .

4 C.

4.1 Capacity Assignment

OD flow F_{ij} , p_{ij} T :

$$\min \sum_{(i,j)} p_{ij} C_{ij} \quad \text{s.t.} \quad \frac{1}{\gamma} \sum_{(i,j)} \frac{F_{ij}}{C_{ij} - F_{ij}} \leq T$$

Lagrangian $L = \sum_{(i,j)} (p_{ij} C_{ij} + \frac{\beta F_{ij}}{\gamma(C_{ij} - F_{ij})})$ $\partial L / \partial C_{ij} = 0$:

$$C_{ij} = F_{ij} + \sqrt{\frac{\beta F_{ij}}{\gamma p_{ij}}}, \quad \beta = \left(\frac{1}{T\gamma} \sum_{(i,j)} \sqrt{p_{ij} F_{ij}} \right)^2$$

: = “flow + ” , $\sqrt{F_{ij}/p_{ij}}$ — , (square-root capacity assignment).

4.2 Concentrator Location

$i \quad j \quad x_{ij} \in \{0, 1\}$:

$$\min \sum_i \sum_j a_{ij} x_{ij} \quad \text{s.t.} \quad \sum_j x_{ij} = 1 \quad (\forall i), \quad \sum_i x_{ij} \leq K_j$$

simplex , $y_j \in \{0, 1\}$ b_j (facility location) .

5



1. : “ (DFDL) , .” . .
2. **Two/three-link M/M/1** closed form: $D'_p = \frac{C_p}{(C_p - x_p)^2} \rightarrow \text{case} \rightarrow$.
3. **FW vs Gradient Projection:** vs , 1 vs 2 , MFDL .
4. **DV vs LS:** . . count-to-infinity .
5. **M/M/1 Little** .

Optimality principle	$\rightarrow$ sink tree
Bellman–Ford	hop DP, $D_i^{h+1} = \min_j [d_{ij} + D_j^h]$, count-to-infinity
Dijkstra	greedy ()
DFDL	,
Frank–Wolfe	$\rightarrow$ MFDL $\rightarrow$ (equal-ratio)
Gradient projection	2 unequal shift,
Two-link	$r \leq C_1 - \sqrt{C_1 C_2}$
Capacity assignment	square-root